

PHYSICS (THEORY) - 33-QUESTION BOARD MASTER PAPER

Strictly Compiled from Sets 55/7/1, 55/7/2, 55/5/2 (2024), 55/5/2(1) (2023)

MAX MARKS: 70

TIME: 3 HOURS

SECTION A (MCQS & ASSERTION-REASON)

1. Two nuclei have mass numbers in the ratio 27:64. The ratio of their nuclear densities is:
(A) 1:1(B) 3:4(C) 9:16(D) 27:64 Multiple Sets
2. A p-n junction diode is forward biased. As a result:
(A) Barrier increases(B) Depletion decreases(C) Depletion increases(D) No change Multiple Sets
3. Part of EM spectrum used in water purifiers to kill germs:
(A) Infrared(B) UV Rays(C) X-rays(D) Microwaves Set 55/7/1
4. Bohr orbit radius r is proportional to:
(A) n (B) n^2 (C) $1/n$ (D) $1/n^2$ Multiple Sets
5. Slope of V_0 vs frequency graph:
(A) h (B) h/e (C) ϕ_0 (D) e/h Set 55/7/1
6. Two parallel wires (2m apart, 3A same dir). Force per unit length:
(A) Attractive(B) Repulsive(C) Zero(D) Constant Set SR5QP
7. Ideal inductor across AC source. Current:
(A) Leads V by $\pi/2$ (B) Lags V by $\pi/2$ (C) Phase(D) Lags V by π Set 55/7/1
8. Amplitude of secondary wavelets is:
(A) Max forward/Zero back(B) Equal both(C) Zero forward(D) Max back Set SR5QP
9. Effective resistance between A and B (Bridge circuit Each resistance 2 ohm):
(A) 2 ohm(B) 6 ohm(C) $8/3$ ohm(D) 3 ohm Set 55/7/1
10. Dimensional formula for μ_0 :
(A) $[MLT^{-2}A^{-2}]$ (B) $[ML^2T^{-1}A^{-1}]$ Set 55/7/2
11. Ser LCR nature before/after resonance:
(A) Cap then Ind(B) Ind then Cap Multiple Sets
12. Metal rod ($L=50\text{cm}$, $v=10\text{m/s}$) in $B=0.4\text{G}$. Induced EMF:
(A) 0.2 mV(B) 2 mV Set 55/7/1
13. **Assertion:** EM waves no medium needed. **Reason:** They are transverse.
14. **Assertion:** Insulators forbade gap is large. **Reason:** Valence e^- tightly bound.
15. **Assertion:** Deflecting torque zero loop perp field. **Reason:** M parallel to B .
16. **Assertion:** Interference not seen if sources infinitely close. **Reason:** beta prop to separation.

SECTION B (2 MARKS)

17. Draw energy band diagrams for n-type and p-type at $T > 0\text{ K}$. Set 55/7/1/2
18. Ratio of drift velocities for wires L & $2L$ and Area A & $2A$ in series. Set 55/7/1/2
19. Calculate Binding Energy per nucleon of Helium nucleus. Set 55/7/2
20. Combo focal length of Convex (10cm) and Concave (15cm) lenses in contact. Set 55/7/1
21. Effect on De Broglie wavelength if accelerating potential increases. Set SR5QP

SECTION C (3 MARKS)

22. Using Huygens verify Snell's law OR Law of Reflection. Set SR5QP
23. Derive H-atom n th orbit radius expression (Bohr Postulates). Set 55/7/1
24. Force between two parallel current conductors. Define 1 Ampere. Set 55/7/1
25. Plot V_0 vs freq. Identify slope and work function from graph. Set 55/7/2
26. Mutual Inductance of two concentric coaxial circular coils. Set 55/7/1

27. Methods to increase MCG sensitivity. MCG to Voltmeter conversion.

Set 55/7/1

28. Potential Energy of 2-charge system in as external field E.

Set 55/7/2

SECTION D (CASE STUDY - 4 MARKS)

29. **Drift Velocity:** (i) Relaxation time vs Temp (ii) Derive $J = ne v_d$ (iii) Define Mobility.

30. **Lens Maker's:** (i) Focal length in water (ii) Multiple images (iii) f for $r_1=10$, $r_2=15$, $n=1.5$.

SECTION E (5 MARKS)

31. (a) Ray diagram for Telescope/Microscope & derive Magnification OR (b) Lens Maker Formula derivation.

32. (a) Working of p-n junction Full Wave Rectifier OR (b) MCG principle, working and radial field.

33. (a) Parallel Plate Cap with dielectric slab OR (b) Potential due to dipole on axial/equatorial line.